

Name
Class
R. No.

• Mathematics
• IIT JEE
• Test 3(July 14 , 2010)
• Functions - Domain of Definition
• Maximum Marks 35
• Time 35 minutes

Instructions

1. The test contains 3 write the answers type questions.
2. The marking scheme will include marks for partially correct answers also.
3. Space is provided after each question for rough work.

Q1: Find the domains of definition of the following functions:

i) $y = \frac{2x-3}{\sqrt{x^2+2x+3}}$

5 Marks

ii) $y = \sqrt{\sin x - 1}$

5 Marks

iii) $y = \frac{1}{\sqrt{|x|-x}}$

5 Marks

Space for rough work:

Q3: Find the range of the following functions

i) $y = \frac{1}{2 - \cos 3x}$

7 Marks

ii) $y = \frac{x}{1+x^2}$

8 Marks

Space for rough work

Q3: Given the function

$$f(x) = \begin{cases} 3^{-x} - 1 & , -1 \leq x < 0 \\ \tan\left(\frac{x}{2}\right) & , 0 \leq x < \pi \\ \frac{x}{(x^2 - 2)} & , \pi \leq x \leq 6 \end{cases}$$

Find $f(-1)$, $f\left(\frac{\pi}{2}\right)$, $f\left(\frac{2\pi}{3}\right)$, $f(4)$, $f(6)$

5 Marks

Space for rough work:
